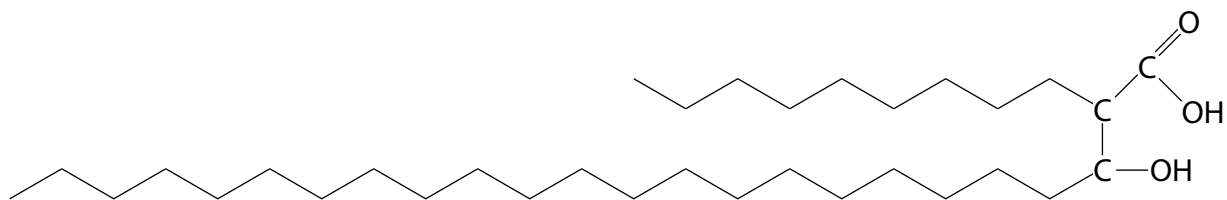


6 Tuberculosis is a disease caused by infection with *Mycobacterium tuberculosis*.

The cell wall of *M. tuberculosis* is different from most other types of bacteria as it contains different types of mycolic acid.

Mycolic acid protects the bacteria from lysozyme action, dehydration and polar (hydrophilic) antibiotics.

The diagram shows part of one type of mycolic acid.



(a) What type of molecule is mycolic acid?

(1)

- ☐ **A** carbohydrate
- ☐ **B** fatty acid
- ☐ **C** nucleic acid
- ☐ **D** polypeptide

- (b) Treatment for tuberculosis involves the use of different types of antibiotic, taken over several months.

The table gives details of four antibiotics used to treat tuberculosis.

Antibiotic	Mode of action
E	disrupts the formation of the cell wall
J	inhibits the synthesis of mycolic acids
P	activates an enzyme that inhibits the synthesis of fatty acids
R	binds to the active site of RNA polymerase

- (i) Suggest why antibiotic E is effective only when *M. tuberculosis* bacteria are dividing.

(1)

.....

.....

.....

- (ii) Explain why antibiotic J could result in an increase in antigen presentation by macrophages.

(2)

.....

.....

.....

.....

.....

.....

(iii) Explain how antibiotic P could affect *M. tuberculosis*.

(3)

(iv) Explain how antibiotic R could affect *M. tuberculosis*.

(3)

- (v) Explain why taking a combination of antibiotics for several months could increase the resistance of *M. tuberculosis* to these antibiotics.

(2)

(Total for Question 6 = 12 marks)
